MIL-E-1/1537B
10 March 1976
SUPERSEDING
MIL-E-1/1537A
9 July 1970

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, KLYSTRON

TYPE 417C

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

Ⓑ DESCRIPTION: Reflex oscillator, integral cavity, adjustable vernier tuning

ABSOLUTE RATINGS:

Parameter:	Ef	Ers	Irs	Prs	Ec	Ic	Er	Ehk	Barometric pressure, reduced	Temperature	
										Cathode seal	Body
Unit:	V	Vdc	mAdc	W	Vdc	mAdc	Vdc	Vdc	mmHg	°C	°C
Maximum:	6.9	1,250	60	75	+50	25	-750	±45	508	100	120
Minimum:	6.0	---	---	---	-200	---	-10	---	---	---	---

PHYSICAL CHARACTERISTICS:

Dimensions: See figure 1

Cathode: Unipotential

Cooling: Forced air (see note 1)

Mounting position: Any

Base: Shortened medium shell octal, 8 pin, low-loss phenolic.

Cap: C1-1

Output: Two output terminals, SKL coaxial fittings, mates with UG-275/U, UG-131/U, or equal.

Tuning: Mechanical vernier tuning lever supplied.

TEST CONDITIONS:

Parameter:	Ef	Ers	Temperature		F
			Cathode seal	Body	
Unit:	V	Vdc	°C	°C	MHz
	6.3	500	---	---	F1 and F3
			(See note 1)		

FREQUENCY		
F	MHz	±
F1	2,660	1%
F2	3,000	1%
F3	3,310	1%

GENERAL:

Qualification - Required

Ⓑ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 1	2					
Ⓑ 4289	Heater current (nonoperating)	3	Ef = 6.3 V	If	0.90	1.10	A
Ⓑ 4270	Heater-cathode leakage	3	Ehk = ±45 Vdc	Ihk	0	150	µAdc
4229	Total reflector current	3, 4	Ec/Irs = 40 mAdc; Er = -60 to -150 Vdc	Ir	0	5	µAdc
1261	Electrode voltage (1) (grid)	3, 4b	Ec/Irs = 20 mAdc	Ec	0	25	Vdc
1256	Electrode current (1) (grid)	3, 4b	Ec/Irs = 20 mAdc	Ic	0	7.5	mAdc
1261	Electrode voltage (2) (grid)	3, 4b	Ers = 1,000 Vdc; Irs = 50 mAdc	Ec	0	50	Vdc
1256	Electrode current (2) (positive grid)	3, 4b	Ers = 1,000 Vdc; Irs = 50 mAdc	Ic	0	25	mAdc
4250	Power output	-	Irs = 25 mAdc (max); Er = -10 to -300 Vdc; ZL = 50 ohms ±10%	Po	50	---	mW
	Quality conformance inspection, part 2						
4027	Temperature coefficient	5	F2; Irs = 25 mAdc (max); Er = -10 to -300 Vdc; ZL = 50 ohms ±10%	ΔF/ΔT	---	120	kHz/°C
Ⓑ 1105	Permanence of marking	-		---	---	---	---
	Quality conformance inspection, part 3						
---	Life test	4b	Group C; Ec/Irs = 25 mAdc	t	500	---	hrs
---	Life-test end point:						
4250	Power output	-	Irs = 25 mAdc (max); Er = -10 to -300 Vdc; ZL = 50 ohms ±10%	Po	35	---	mW
Ⓑ 1031	High-frequency vibration	7		ΔF	---	100	kHz
Ⓑ 1031	Low-frequency vibration	6, 7	Er = -10 to -300 Vdc	BW	---	1.5	MHz
Ⓑ 4223	Mechanical tuning range (vernier)	7	F1 F3	--- ---	160 100	--- ---	MHz MHz
Ⓑ 4223	Mechanical tuning range	7	Low frequency High frequency	F F	--- 3,310	2,660 ---	MHz MHz
Ⓑ 4223	Mechanical tuning fatigue (1)	7, 8		---	15	---	Cycles
Ⓑ 4223	Mechanical tuning fatigue (2)	7, 9		---	165	---	Cycles

NOTES:

1. Sufficient cooling shall be distributed over the entire tube so that maximum temperature ratings are not exceeded. Recommended cooling at sea level, when operating at maximum power input (Prs), is 35 cfm. At no time during test should tuning struts be loosened to cause grids to be completely drawn together. This may damage tube.
2. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.
3. This test shall be performed at the conclusion of the holding period.

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NOTES: -Continued

4. Total reflector current.

   (a) Apply heater voltage for 1 minute, then apply other voltages for 2 minutes, and read the total reflector current at the end of this 2-minute period.

   (b) The TUT shall not be oscillating during test. If the TUT oscillates with the specified voltages applied, detune the resonator until oscillation ceases.

5. The initial frequency shall be allowed to stabilize by operation at practically constant ambient temperature for at least 10 minutes. The temperature range for the test shall be over the range of -10° to +40°C. The rate of change of ambient temperature shall not be greater than 2°C per minute. During the test, the TUT shall be operated in a rectangular enclosure that has a base at least 1-foot square. The enclosure shall be at least 20-inches high. An airblast, as specified in note 1, shall be directed at the tube. The rate of airflow past the TUT shall be maintained as nearly constant as practical. The air within the enclosure shall also be circulated.

6. The total frequency bandwidth including any permanent change in frequency, shall not exceed the value specified. Bandwidth determination shall be made during the test.

7. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with an combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

8. The number of full specified frequency range flex cycles shall not be less than the amount specified.

9. The number of fine adjustment flex cycles shall not be less than the amount specified.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MU
Air Force - 70, 80
DSA - ES

User activities:
Navy - AS, OS, MC, CG
Air Force - 11

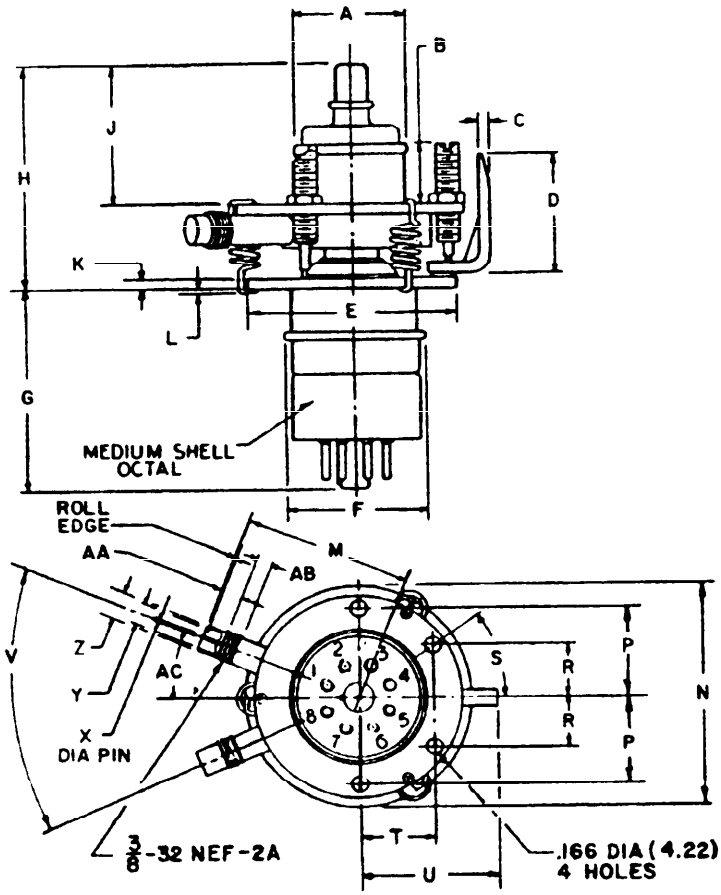
Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-3074)

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 1 (see note 2)		
K	.110 (2.79)	.140 (3.56)
S	34° 0'	36°0'
V	41° 0'	49°0'
X	.048 (1.22)	.052 (1.32)
Z	.279 (7.09)	.283 (7.19)
AA	.234 (5.94)	.266 (6.76)
Quality conformance inspection, part 2		
A		1.250 (31.75)
B		.950 (24.13)
E		2.220 (56.39)
F		1.500 (38.10)
G		2.325 (59.06)
H		2.690 (68.33)
J		1.375 (34.93)
M		1.750 (44.45)
N		2.625 (66.68)
Nominal dimensions		
C	.125 (3.18)	
D	1.280 (32 51)	
L	.062 (1.57)	
P	.953 (24.21)	
R	.547 (13.89)	
T	.781 (19.84)	
U	1.530 (38.86)	
Y	.230 (5.84)	
AB	.156 (3.96)	
AC	22.5°	



BASE CONNECTIONS

1. I.C. (Internal connection)
2. Heater
3. I.C.
4. I. C.
5. Control electrode
6. I.C.
7. Heater
8. Cathode

Shell-resonator
Cap-reflector

FIGURE 1. Outline drawing of electron tube type 417C.

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